



Alexander Miale



Personal Statement

I believe the outcome of every action should have a positive impact on the lives of others. I also believe technology can be used for this, although it might not always be necessary. Because of this, my preferred line of work lies at the intersection of technology, health, and policy. I have the technical skills to contribute to most modern tech stacks, specifically for data analytics in Python and SQL or NoSQL databases, and the soft skills to guide a project and ensure its impact remains positive. For these reasons, I want to work for the betterment of people and society, especially those who are underprivileged and oppressed.



Work Experience

Research Assistant - C++, JUCE, RTNeural, CMake
@ Northeastern University (August 2025 - April 2026)

- Conducted research with Professor Victor Zappi to **develop modern machine learning solutions** for **musical AI models** that run on **accelerated processors** to be used in **real-time** scenarios
- Experimented with both **grey box** (differentiable digital signal processing) and **black box** (neural networks) solutions
- Utilized **RTNeural** to **load and run** model weights from **JSON files**
- Built **two audio effect plugins** for digital audio workstations
- Emulated the **LA-2A leveling compressor** grey-box model and a **TS9 distortion pedal** black box model
- Awarded the **PEAK Fellowship: The Summit** for the spring semester to continue the research and present findings at **RISE**

Crew Member

@ Trader Joe's (August - December 2025)

- **Upheld the values** of Trader Joe's to ensure a perfect experience for the customers
- Aided customers with the checkout processes and **created a welcoming environment** at the registers and demo stations
- Stocked shelves to **keep the store presentable** and allow customers to find all of their favorite items
- **Assisted customers** with any issues or questions they had



Technical Projects - Spring 2026

EPV and xG Models - PyTorch, Matplotlib, R, ggplot2

- Created two machine learning models that **predict the likelihood of a goal** being scored given **different player position and situational characteristics**
- Utilized **SkillCorner's open data**, containing 10 matches of Australian A-League **tracking and dynamic event data**
- Each model required different pieces of data, so two **different preprocessing pipelines** were created in **Python**
- Implemented a **neural network using PyTorch** in Python for the EPV model, and a **Bayesian logistic regression Model in R** for xG
- Finished with **0.99 ROC/AUC** for EPV and **0.73 ROC/AUC** for xG



Contact & Info

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Location: Kennebunk, Maine



Education

School: Northeastern University

Degree: BS in Data Science,
Minor in Music Technology

GPA: 3.8/4.0, Magna Cum Laude

Graduated: April 2026

Relevant Courses: Computer Science Foundations I and II, Machine Learning I and II, Artificial Intelligence, Information Presentation and Visualization, Large-Scale Storage/Retrieval, Practical Neural Networks, Human/Computer Interaction, Calculus I and II, Probability and Statistics, Linear Algebra



Technical Skills

Languages: Java, Python, R, C++

Databases: SQL, Redis, Neo4j, MongoDB, Splunk

Other Technologies: NumPy, Pandas, scikit-learn, Matplotlib, TensorFlow, Torch, Git, Docker, Tableau, ggplot2, CMake